

GR818 Digital Temperature Controller	
Operation Instruction V3. 02	
Thank you for using Cautoni-Swastik GR818 digital temperature controller	
This instruction describes product function, features and proper usage	
Before using this controller, please pay special attention to issues below:	
• User should master enough electric knowledge • User must read and understand this instruction well for right usage • Please consider applicability to system, machine and equipment • Please note and observe the prohibition of this product • The examples in this instruction or other data are only for user reference. No guarantee of a certain action. • When using this controller with other products, please confirm whether it is in conformity with the relevant specifications, principles, etc.. • Please keep this instruction well to make it available easily anytime	

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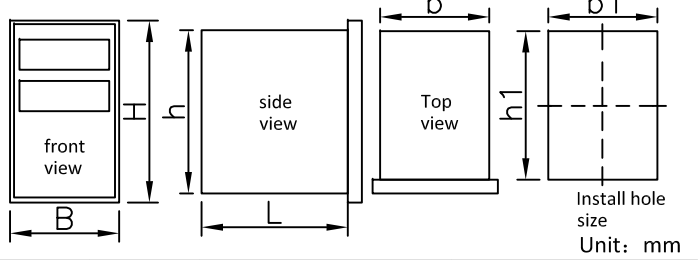
Warning	
①	To avoid electric shock, DON'T touch the AC power supply terminals after power on
②	Before power on, please confirm voltage conformity in range of AC 85~265V, to avoid damage to controller
③	Never remove, retrofit or repair the product or contact any internal components
④	If the output relay exceeds its life span, the contacts may melt and burn.
⑤	Tighten screw with 0.74-0.90Nm torque, as loose screws may cause fire
⑥	Use appropriate fuse to ensure power supply line and input / output line to prevent current impact
⑦	Don't use the controller in occasion of flammable, explosive gas, discharge of steam
Note	
①	For heat radiation, leave space around the controller and do not jam the ventilation holes in the controller.
②	Keep enough space between the controller and equipment which generates high frequency and surge
③	Connect wires correctly
④	Use the controller under rated load and power supply
⑤	Use standard grade of alcohol to clean the controller; don't use paint thinner or similar chemicals
⑥	Read and understand this instruction carefully before using the controller
⑦	Don't use the controller in case the front panel peels off or breaks

A3 Electric specification	
rated voltage	85V-265V AC , 50/60HZ
power comsumption	≤5VA
working environment	Ambient temperature: 0C ~50C Relative humidity: 35%~85% (no condensation)
Storage temperature	-25℃-65℃（Avoid ice or dew）
Resolution	1℃, 0.1℃（adjustable）
Wiring method	terminals
accuracy	±0.5%FS
Memory protection	Non-volatile memory（write times: 1,000,000）
installation environment	Installation type II, pollution grade 2（IEC61010-1）
relay output	relay contacts AC220V/DC30V, 3A（Resistive load）
logic level output	ON:DC12V; OFF: below DC0.5V; Max current: 30mA; Load resistance≥1K.

A4 Model selection

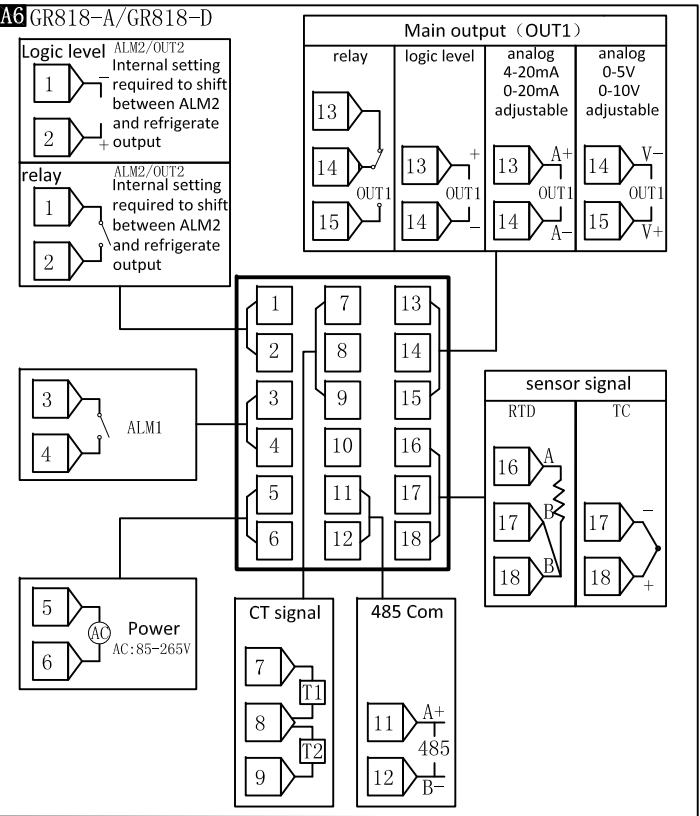
Model code explation	
GR818 — □□□□□□□ ①②③④⑤⑥ — □□□□ ⑦	
① Panel size	A: 48*48 B: 48*96 D: 72*72 E: 96*96
② Input signal	T: Thermocouple/RTD input L: Voltage/current signal input(0-10V/0-20mA)
③ Main output	1: relay 2: logic level 3: analog（0-5V/0-10V/4-20mA） 4: SCR phase shift 5: SCR zero cross
④ Alarm output	1: 1 relay alarm output 2: 2 relay alarm outputs 3: 1 logic level alarm output and 1 relay alarm output
⑤ Strengthened function	0: No strengthened function 1: RS485 communication 2: current measure 3: RS485 communication+current measure 4: analog input channel2(0-10V/0-20mA)
⑥ Expension input output	00: no expansion 22: 2 input 2 output 40: 4 input 04: 4 output
⑦ Function code	

A5 Outer dimension and install hole size

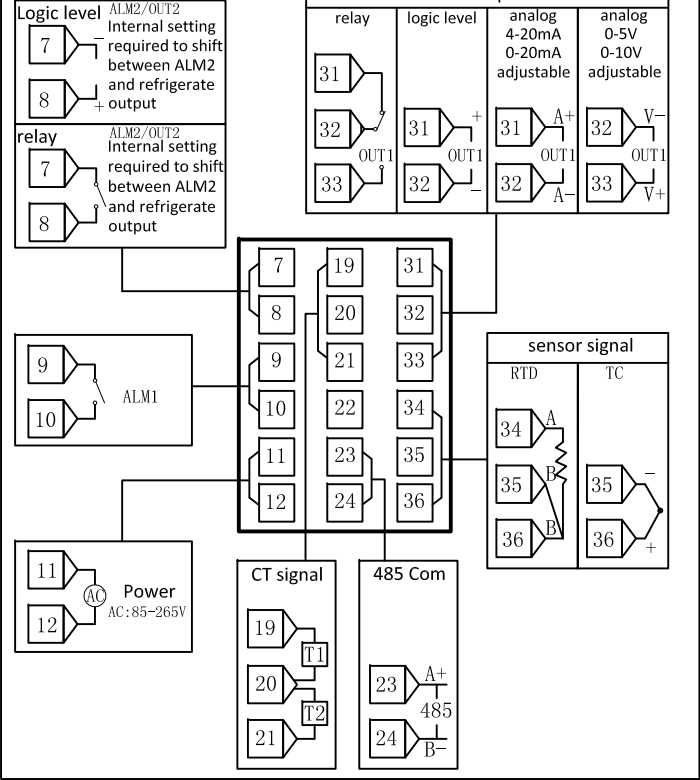


Model	Panel siz H×B	case size h×b×L	install hole size h1×b1
GR818-A	48×48	44.56×44.56×79.00	(44.56+1)×(44.56+1)
GR818-B	96×48	91.00×44.56×79.00	(91.00+1)×(44.56+1)
GR818-D	72×72	67.00×66.80×79.00	(67.00+1)×(66.80+1)
GR818-E	96×96	91.00×90.50×79.00	(91.00+1)×(90.50+1)

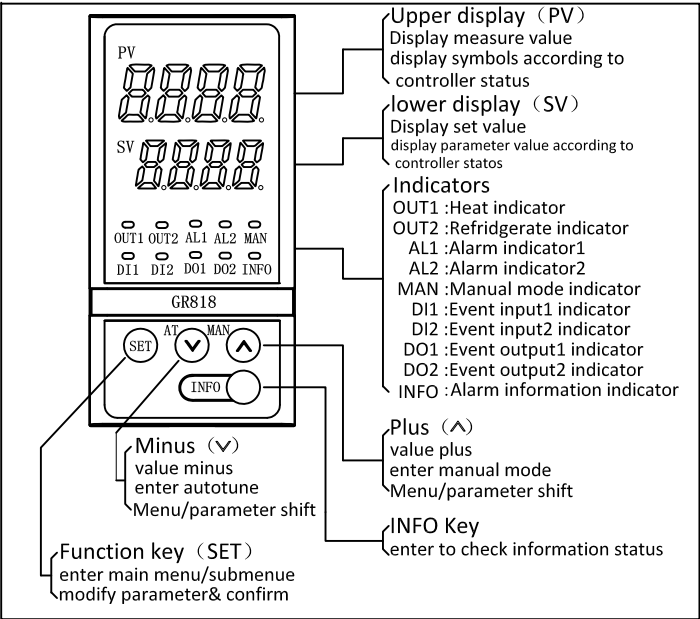
Wiring



B1 GR818-B/GR818-E



B2 Operation panel explanation



Alarm

B3 ALM alarm parameter define			
Code	alarm name	Code	alarm name
0	No alarm	1	upper limit
2	lower limit	3	upper/lower limit
4	upper/lower limit range	5	upper limit (keep)
6	lower limit(keep)	7	upper/lower limit (keep)
8	upper/lower limit range (keep)	9	absolute value upper limit
10	absolute value upper limit	11	absolute value upper/lower limit
12	absolute value upper/lower range	13	absolute value upper limit (keep)
14	absolute vakeepl	15	absolute value upper/lower limit (keep)
16	absolute value lower limit range (keep)	17	upper limit reture difference
18	lower limit reture difference	19	absolute value upper limit reture difference
20	absolute value lower limit reture difference	21	upper limte reture difference (keep)
22	lower limit return difference (keep)	23	absolute value upper limte reture difference(keep)
24	absolute vale lower limit return difference(keep)		

B4 ALM alarm output function define			
Code	alarm name	Code	alarm name
0	no configure	1	current measure NO/NC status
2	current measure NO status	3	current measure NC status

Code	alarm name	Code	alarm name
4	alarm output1（ALM1）	5	alarm output2（ALM2）
6	alarm output 3（ALM3）	7	heat output
8	refrigerate output	9	manual symbol
10	information alarm symbol		

B5 Input type

temperature input	thermocouple	input	Set code	temperature range
		K		-200℃—1200℃
		E		-200℃—650℃
		J		-200℃—850℃
		S		-50℃—1700℃
		B		0℃—1800℃
		N		-200℃—1300℃
		R		-50℃—1700℃
		T		-200℃—400℃
		W3-25		0℃—2300℃
		W5-26		0℃—2300℃
	RTD	PT100		-200℃—850℃
		Cu50		-50℃—150℃

B6 control method and sensor type selection

control method	Main menu	submenu	set code	code define
				PID control
				two-way PID control (heat/refridgerate)
sensor type				Step control
			TC	K、E、J、S
			TC	B、N、R、T
			RTD	W3-25、W3-26
			RTD	PT100、Cu50

B7 information status explanation

INFO information status table			
Code	Code define	Code	Code define
F. 100	currnet value measured	F. 101	manual output value
F. 102	maintain output value	F. 103	temperature control output value
F. 104	cold end temperature (room temperature)	F. 200	total curent value of channel 1
F. 201	total current value of channel2	F. 202	Min current per single wire
F. 203	temperature resistor value (RTD)	F. 204	temperature voltage value(TC)
F. 300	MODBUS Baud rate	F. 301	MODBUS station number
F. 302	RS485 communication status	F. 303	USB port communication status
F. 304	Phase shift board communication status	F. 400	controller working days
F. 401	relay output times	F. 900	producing date: year
F. 901	producing date: month	F. 902	producing date: day
F. 903	menu table version	F. 904	system software version
F. 905	user customize version		

B8 Error display explanation

Error code table			
Error display	code explanation	Troubleshooting	Remark
Err0	exceed AD input upper limit	check if thermocouple is disconnected	PV display tube flashes
Err1	exceed AD input lower limit	check if analog input is connected reversedly	
Err2	RTD Break	check if RTD is disconnected	
Err3	exceed measure range upper limit	check if the used sensor type is in accordance with setting	
Err4	exceed measure range lower limit	check if the used sensor type is in accordance with setting	
Err5	exceed set value upper limit	check if the SV upper limit is too small	INFO indicator flashes, press INFO key to checkdetails
Err6	exceed set value lower limit	check if the SV upper limit is too big	
Err7	room temperature error	Controller malfunction, please contact supplier for service.	
Err8	thermocouple connect reversed	check if thermocouple is connected reversedly	
F. 001	control cycle set vaule conflict	set reasonable control cycle	
F. 002	output breaks	check if heating circuitbreaks	INFO indicator flashes, press INFO key to checkdetails
F. 003	output adheres	check if heat circuit has unnormal NC parts	
F. 004	relay output reaches set times	change controller	
F. 005	working temperature exceed	check if temperature in cabinet is too high	

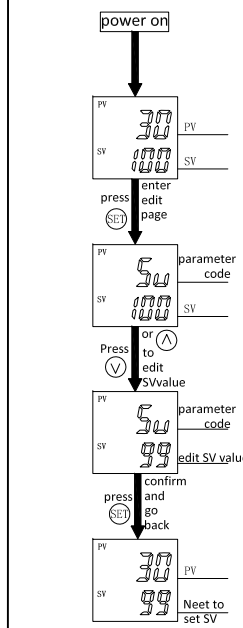
C Menu display explanation

main menu	submenu	menu define	Add
1 Pid (PID) basic control parameter	1-1 P	P Proportional band	W211
	1-2 I	I Integral time	W212
	1-3 d	D Derivative time	W213
	1-4 IT	IT Overshooting restrain	W214
	1-5 HT	HT heat cycle	W215
	1-6 CT	CT refrigerate cycle	W216
	1-7 $HHYS$	HHYS heat return difference	W204
	1-8 AT	Autotune switch	W206
2 $Pid2$ (PID2) main control parameter	2-1 OBJ	OBJ control model	W210
	2-2 PC	PC refrigerate P	W217
	2-3 IC	IC refrigerate I	W225
	2-4 DC	DC refrigerate D	W226
	2-5 CPK	CPK Refrigerate power coefficient	W224
	2-6 $RUCY$	RUCY valve stroke time	W151
	2-7 BD	BD dead zone value	W203
	2-8 $INOS$	INOS Overshoot suppression factor	W208
	2-9 LEV	LEV Control intensity coefficient	W209
	2-10 CPO	CPO Proportional refrigeration displacement	W218
	2-11 $T-ON$	T-ON Min boot time	W219
	2-12 $T-OF$	T-OF Min shutdown time	W220
	2-13 $OMAX$	OMAX max output value limit	W221
	2-14 $OMIN$	OMIN min output value limit	W222
	2-15 $DMAX$	DMAX Output variationlimit	W223
3 $TEMP$ (TEMP) temperature measure	3-1 $P-N1$	P-N1 Control method	W201
	3-2 $P-N2$	P-N2 input type *1	W133
	3-3 TR	TR input compensation *1	W136
	3-4 $P-DP$	P-DP display accuracy *1	W138
	3-5 $P-CF$	P-CF Fahrenheit *1	W137
	3-6 $P-N2$	P-N2 Input type *2	W150
	3-7 $D-MD$	D-MD compensation type *2	W118
	3-8 $D-iH$	D-iH filter input high point *2	W119
	3-9 $D-iL$	D-iL filter input low point *2	W120
	3-10 $D-oH$	D-oH filter output high point *2	W121
	3-11 $D-oL$	D-oL filter onput low point *2	W122
	3-12 $P-SH$	P-SH set upper limit	W134
	3-13 $P-SL$	P-SL set lower limit	W135
	3-14 FiT	FiT temperature process	
	3-14-1 $C-MD$	(C-MD) temperature filter type	W115
	3-14-2 $COP1$	(COP1) filter parameter1	W116
	3-14-3 $COP2$	(COP2) filter parameter2	W117
	3-14-4 $D-MD$	(D-MD) compensation type	W118
4 ALM (ALM) alarm configure	4-1 $ALM1$	ALM1 alarm type1	W180
	4-2 $AH1$	AH1 alarm upper limit 1	W181
	4-3 $AL1$	AL1 alarm lower limit1	W182
	4-4 $AOP1$	AOP1 alarm option1	W183
	4-5 $ALM2$	ALM2 alarm type 2	W184
	4-6 $AH2$	AH2 alarm upper limit 2	W185
	4-7 $AL2$	AL2 alarm lower limit2	W186
	4-8 $AOP2$	AOP2 alarm option2	W187
	4-9 $ALM3$	ALM3 alarm type3	W188
	4-10 $AH3$	AH3 alarm upper limit3	W189
	4-11 $AL3$	AL3 alarm lower limit 3	W190
	4-12 $AOP3$	AOP3 alarm option 3	W191

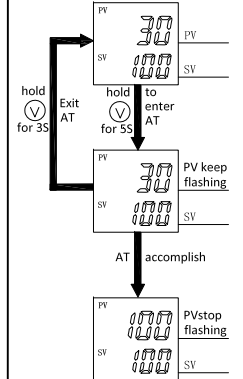
D Operation guide

main menu	submenu	menu define	Add
5 SVC (SVC) multi stage temperature control (programmable controller parameter)	5-1 $MODE$	MODE multistage control mode	W142
	5-2 $S-ID$	S-ID multistage control number	W141
	5-3 SV	SV set temperature	W131
	5-4 $T-ST$	T-ST Heating time	W139
	5-5 $J-SV$	J-SV Energy saving temperature	W512
	5-6 $J-TM$	J-TM Energy saving delay time	W502
	6-1 $AOMD$	AOMD analog output mode	W126
	6-2 $AOPZ$	AOPZ analog output configure	W127
	6-3 AOH	AOH Analog output transmission high point	W128
	6-4 AOL	AOL Analog output transmission low point	W129
	6-5 $M-SW$	M-SW control mode (manual/auto)	W171
6 MV (MV) output control	6-6 MAN	MAN manual output value	W170
	6-7 $E-MV$	E-MV error output value	W172
	6-8 $K-SW$	K-SW maintain switch	W174
	6-9 $K-TM$	K-TM maintain judge time	W175
	6-10 $K-PC$	K-PC maintain judge deviation range	W176
	6-11 $K-MV$	K-MV maintain valve *3	W173
	6-12 $K-CT$	K-CT maintain count value *3	W177
	7-1 SET	SET total current measure switch	W270
	7-2 ONE	ONE min single current	W263
	7-3 $CH1$	CH1 current save-channel 1	W264
	7-4 $CH2$	CH2 current save-channel 2	W265
7 $CURR$ (CURR) transformer current	7-5 DSP	DSP current display (C-MD) current measure mode	W262
	7-5-1 $C-SW$	(C-SW) current display switch	W937
	7-5-2 $C-FM$	(C-FM) transformer denominator	W260
	7-5-3 $C-FZ$	(C-FZ) transformer numerator	W261
	7-5-4 $CT1$	(CT1) real time total current *3	W253
	7-5-5 $CT1$	(CT1) real time current1 *3	W254
	7-5-6 $CT2$	(CT2) real time current2 *3	W255
	7-5-7 KP	(KP) keep total current(2sec) *3	W257
	7-5-8 $KP1$	(KP1) keep current1 *3	W258
	7-5-9 $KP2$	(KP2) keep current2 *3	W259
	7-5-10 $KEEP$	(KEEP) current(long term)*3	W256
8 SYS (SYS) system configure	8-1 $BAUD$	BAUD MODBUS baud rate	W900
	8-2 $PCOL$	PCOL MODBUS station No.	W901
	8-3 BRI	BRI brightness	W936
	8-4 $R-SW$	R-SW relay detect switch *4	W123
	8-5 $R-JS$	R-JS relay times *4	W125
	8-6 RUN	RUN running time *3	W908
	8-7 $INFO$	INFO system information	
	8-7-1 $HD1$	(HD1) Hardware type1 *3	W919
	8-7-2 $HD2$	(HD2) Hardware type2 *3	W920
	8-7-3 $HD3$	(HD3) Hardware type3 *3	W921
	8-7-4 $HD4$	(HD4) Hardware type4 *3	W922
9 DEF (DEF) parameter default	9-7-5 $YEAR$	(YEAR) producing year *3	W909
	8-7-6 MON	(MON) producing month *3	W910
	8-7-7 DAY	(DAY) producing day *3	W911
	8-7-8 $HOURL$	(HOURL) producing hour *3	W912
	8-7-9 MIN	(MIN) producing min *3	W913
	8-7-10 SEC	(SEC) producing sec *3	W914
	8-7-11 PID	(PID) control base version *3	W924
	10 $LOCK$	(LOCK) 0-lock data 1-unlock basic data 8888-unlock all data	
	label	**1**display with temperature input **3**parameter read only	
	Modbus	Format:data bit 8, stop bit 1, even parity, all in decimal system	
		Common address:PV-W130,SV-W131,Heah indicator-B224,Autotune-B3296	

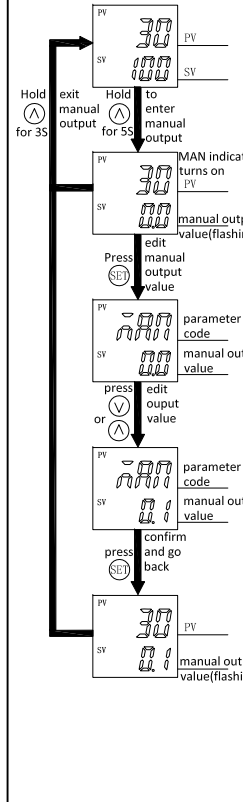
Temperature setting



Autotune(AT)



Manual output(MAN)



Check information stauts (INFO)

